Cancer and aging: from the kinetics of biological parameters to the kinetics of cancer incidence and mortality.

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Epidemiologic and biological data strongly support the existence of a strict link between cancer and aging. In spite of the relevance of the problem, there were numerous pitfalls in epidemiologic investigation until a few years ago. An apparent decrease of cancer incidence in old age was revealed to be a misconception based on lack of sufficient appreciation for changing population size. But not all problems are solved by using age-specific cancer incidence, as recently stressed by some authors. At very advanced ages a slowing of the rate of increase of age-specific cancer incidence is clearly demonstrated. These findings apparently clash with the majority of biological data and suggest that some mechanism may develop at advanced ages capable of decreasing cancer susceptibility. In this paper, it will be shown that just a slowing-down kinetics is predicted for cancer incidence by using a mathematical model of mortality kinetics recently proposed in the gerontologic field. The slowing of the increasing rate or even a decreasing trend of cancer incidence of an aging population is compatible with a continuously accelerating pace of loss of physiological capacity of the single subjects, as with advancing age there is a selection of individuals with better physiological functions.